Project 2: ChatGPT and Mechanics
Chapter 3, Problem 3 (Hamster + Armadillo-in-a-Ball in a Rolling Wheel)

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Project 2: ChatGPT and Mechanics.

Choose one of the homework problems from Chapter 3 involving a hamster in a ball. Use a generative program like ChatGPT to:

(a) derive the equations of motion,

(b) write a Python program that generates the numerical solution,

(c) run the program and plot the solution.

The generative AI will most likely not be able to solve the problem directly. If that assumption is true, observe how much "directing" is required.

Problem statement (Chapter 3, Problem 3)

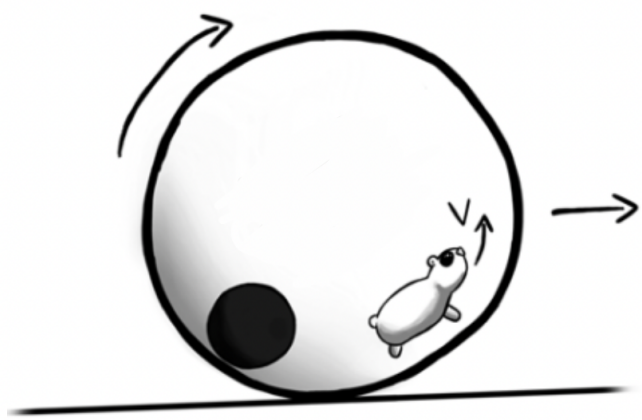

Figure 1: Problem setup (a rolling wheel containing a hamster and an armadillo rolled in a ball).

A vertical wheel rolls freely without slipping on a horizontal table. The moment of inertia of the wheel is I, mass is M, and radius is R. In the wheel, there is a hamster running with a constant speed V. The hamster has mass m_1, and it is separated from the wheel's edge by height h_1, so the hamster's center of mass is always $R - h_1$ away from the wheel's center. Now there is an armadillo rolled in a ball, rolling freely without slipping inside the wheel. The masses of the hamster and armadillo are m_1 and m_2; the distances from the edge of the wheel to their centers of mass are h_1 and h_2, respectively; and the velocity of the hamster is V (fixed). Since the tracks are parallel, when the hamster and armadillo meet, they pass through each other without interaction.

Find the configuration manifold of the problem and write the Euler–Lagrange equations. Find the energy E. Is the energy conserved, and is it equal to the kinetic+potential energy? Are there any equilibria for some values of parameters? If yes, find them and describe the motion about the equilibria (stable or unstable). You don't need to do stability computations here. In this problem, you are only asked to consider the equilibrium where both the hamster and armadillo are down.

(a) Deriving the equations of motion

Problem set-up and coordinates

The system consists of:

- A vertical wheel of radius R, mass M, moment of inertia I, rolling without slipping on a horizontal table.
- Inside the wheel there is:
  - a hamster of mass m_H, running with constant speed V on a circular track of radius $R - h$;
  - an armadillo in a ball of mass m_A and moment of inertia I_A, whose centre also moves on a circle of radius $R - h$.
- The ball rolls without slipping on the inner surface of the wheel.

Following the book, I choose two generalized coordinates:

- $\varphi(t)$: rotation angle of the wheel (and of the hamster track), measured from the downward vertical;
- $\theta(t)$: angle of the vector from the wheel centre to the armadillo's centre, also measured from the downward vertical.

The configuration space is therefore

$$Q = S^1 \times S^1.$$

The hamster runs with constant angular speed

$$\Omega_H = \frac{V}{R - h}$$

relative to the wheel, so its absolute angle is $\Omega_H t + \varphi$.

Positions and velocities

With the wheel centre at $\mathbf{x}_B = (R\varphi, 0)$, the book gives the positions:

$$\begin{aligned}
\mathbf{x}_H &= \big(R\varphi + (R-h)\sin(\Omega_H t + \varphi),\ -(R-h)\cos(\Omega_H t + \varphi)\big),\\
\mathbf{x}_A &= \big(R\varphi + (R-h)\sin\theta,\ -(R-h)\cos\theta\big).
\end{aligned}$$

Differentiating in time leads to the velocities:

$$\begin{aligned}
\dot{\mathbf{x}}_B &= (R\dot{\varphi}, 0),\\
\dot{\mathbf{x}}_H &= \Big(R\dot{\varphi} + (R-h)\cos(\Omega_H t + \varphi)(\Omega_H + \dot{\varphi}),\ (R-h)\sin(\Omega_H t + \varphi)(\Omega_H + \dot{\varphi})\Big),\\
\dot{\mathbf{x}}_A &= \Big(R\dot{\varphi} + (R-h)\cos\theta\,\dot{\theta},\ (R-h)\sin\theta\,\dot{\theta}\Big).
\end{aligned}$$

For the armadillo ball, denote by $\psi(t)$ the rotation angle of the ball about its own centre. No slip between the ball and the inner wall (radius $R-h$) gives the relation (book Eq. (A.15)):

$$h\dot{\psi} + (R-h)\dot{\theta} = R\dot{\varphi} \qquad \Rightarrow \qquad \dot{\psi} = \frac{R}{h}\dot{\varphi} - \frac{R-h}{h}\dot{\theta}.$$

Kinetic and potential energies

Using the expressions above, the kinetic energies are:

- Wheel:

$$T_B = \frac{1}{2}(MR^2 + I)\dot{\varphi}^2.$$

- Hamster:

$$T_H = \frac{1}{2}m_H\left[R^2\dot{\varphi}^2 + (R-h)^2(\dot{\varphi} + \Omega_H)^2 + 2(R-h)R\dot{\varphi}(\dot{\varphi} + \Omega_H)\cos(\Omega_H t + \varphi)\right].$$

- Armadillo (translation + ball rotation):

$$T_A = \frac{1}{2}m_A\left[R^2\dot{\varphi}^2 + (R-h)^2\dot{\theta}^2 + 2(R-h)R\dot{\varphi}\dot{\theta}\cos\theta\right] + \frac{I_A}{2h^2}\Big(R\dot{\varphi} - (R-h)\dot{\theta}\Big)^2.$$

For the potential energy, taking zero at the wheel centre,

$$V = -m_H g(R-h)\cos(\Omega_H t + \varphi) - m_A g(R-h)\cos\theta.$$

So the Lagrangian in terms of (φ, θ) is

$$L(\varphi, \theta, \dot{\varphi}, \dot{\theta}, t) = T_B + T_H + T_A - V,$$

which matches the book's expression (Eq. (A.16)).

Time-independent Lagrangian

To remove the explicit time dependence coming from the hamster's prescribed motion, I follow the book and introduce

$$q(t) = \varphi(t) + \Omega_H t, \qquad \dot{q} = \dot{\varphi} + \Omega_H \quad \Rightarrow \quad \dot{\varphi} = \dot{q} - \Omega_H.$$

Using q and θ as generalized coordinates, and substituting $\dot{\varphi} = \dot{q} - \Omega_H$ into the energies above leads to a time-independent Lagrangian (book Eq. (A.17)):

$$\begin{aligned} L(q, \theta, \dot{q}, \dot{\theta}) = & \frac{1}{2}(MR^2 + I)(\dot{q} - \Omega_H)^2 \\ & + \frac{1}{2} m_H \left[R^2(\dot{q} - \Omega_H)^2 + (R-h)^2 \dot{q}^2 + 2(R-h)R(\dot{q} - \Omega_H)\dot{q} \cos q \right] \\ & + \frac{1}{2} m_A \left[R^2(\dot{q} - \Omega_H)^2 + (R-h)^2 \dot{\theta}^2 + 2(R-h)R(\dot{q} - \Omega_H)\dot{\theta} \cos \theta \right] \\ & + \frac{I_A}{2h^2} \left(R(\dot{q} - \Omega_H) - (R-h)\dot{\theta} \right)^2 + m_H g (R-h) \cos q + m_A g (R-h) \cos \theta. \end{aligned}$$

Euler–Lagrange equations

The equations of motion for q and θ are

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0, \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0.$$

The book lists the required partial derivatives (Eq. (A.18)); substituting them into the Euler–Lagrange equations yields two coupled nonlinear second-order ODEs for $q(t)$ and $\theta(t)$. I did not fully expand them by hand; instead I used ChatGPT + SymPy to obtain explicit formulas and then integrated them numerically.

(b) Python program for the numerical solution

The key point is that I explicitly encoded the book's time-independent Lagrangian and then used SymPy to construct the Euler–Lagrange equations and solve for $\ddot{q}$ and $\ddot{\theta}$.

```
import numpy as np
import sympy as sp
from scipy.integrate import solve_ivp
import matplotlib.pyplot as plt

# --------------------------
# 1. Parameters
# --------------------------
R = 0.5 # wheel radius (m)
h = 0.10 # distance from inner wall to ball centre (m)
M = 2.0 # wheel mass (kg)
I = 0.5 * M * R**2 # solid cylinder
mH = 0.20 # hamster mass (kg)
mA = 0.30 # armadillo+ball mass (kg)
IA = 0.4 * mA * h**2 # approximate solid sphere
V = 0.5 # hamster speed along the track (m/s)
g = 9.81
```

```python
OmegaH = V / (R - h) # hamster angular speed

# --------------------------
# 2. Symbolic Lagrangian
# --------------------------
q, th = sp.symbols('q th', real=True)
qd, thd = sp.symbols('qd thd', real=True)

L = sp.Rational(1,2) * (M*R**2 + I) * (qd - OmegaH)**2
L += sp.Rational(1,2) * mH * (
        R**2 * (qd - OmegaH)**2
      + (R - h)**2 * qd**2
      + 2 * (R - h) * R * (qd - OmegaH) * qd * sp.cos(q)
)
L += sp.Rational(1,2) * mA * (
        R**2 * (qd - OmegaH)**2
      + (R - h)**2 * thd**2
      + 2 * (R - h) * R * (qd - OmegaH) * thd * sp.cos(th)
)
L += (IA / (2*h**2)) * (R*(qd - OmegaH) - (R - h)*thd)**2
L += mH * g * (R - h) * sp.cos(q) + mA * g * (R - h) * sp.cos(th)

qdd, thdd = sp.symbols('qdd thdd', real=True)

def ddt(expr):
    return (sp.diff(expr, q) * qd +
            sp.diff(expr, th) * thd +
            sp.diff(expr, qd) * qdd +
            sp.diff(expr, thd) * thdd)

dL_dqd = sp.diff(L, qd)
dL_dthd = sp.diff(L, thd)

EL_q = sp.simplify(ddt(dL_dqd) - sp.diff(L, q))
EL_th = sp.simplify(ddt(dL_dthd) - sp.diff(L, th))

sol = sp.solve([EL_q, EL_th], (qdd, thdd), simplify=True)

qdd_expr = sp.simplify(sol[qdd])
thdd_expr = sp.simplify(sol[thdd])

qdd_fun = sp.lambdify((q, th, qd, thd), qdd_expr, 'numpy')
thdd_fun = sp.lambdify((q, th, qd, thd), thdd_expr, 'numpy')

# --------------------------
# 3. ODE system
# --------------------------
def rhs(t, y):
    q_val, th_val, qd_val, thd_val = y
    return [
        qd_val,
        thd_val,
        qdd_fun(q_val, th_val, qd_val, thd_val),
```

```
        thdd_fun(q_val, th_val, qd_val, thd_val)
    ]

# --------------------------
# 4. Numerical integration
# --------------------------
q0, th0 = 0.05, -0.03
qd0, thd0 = 0.0, 0.0
y0 = [q0, th0, qd0, thd0]

t_span = (0.0, 20.0)
t_eval = np.linspace(*t_span, 2000)

sol_ivp = solve_ivp(rhs, t_span, y0, t_eval=t_eval,
                    rtol=1e-8, atol=1e-9)

# --------------------------
# 5. Plotting
# --------------------------
t = sol_ivp.t
q_t = sol_ivp.y[0]
th_t = sol_ivp.y[1]

plt.figure(figsize=(8, 4))
plt.plot(t, q_t, label='hamster angle q(t)')
plt.plot(t, th_t, label='armadillo angle theta(t)')
plt.xlabel('time t (s)')
plt.ylabel('angle (rad)')
plt.title('Hamster and armadillo in a rolling wheel')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

(c) Running the code and interpreting the motion

With the chosen parameter values and small initial displacements, the numerical integration on $t \in [0, 20]$ s produces smooth oscillatory solutions for both generalized coordinates $q(t)$ and $\theta(t)$.

- Both angles oscillate about zero, corresponding to the configuration where the hamster and the armadillo are located at the bottom of the wheel. The two motions occur on different time scales: $q(t)$ exhibits slower oscillations with larger amplitude, while $\theta(t)$ undergoes faster, smaller-amplitude oscillations.
- The difference in frequencies reflects the different effective inertias and constraints of the two subsystems. The motions are dynamically coupled through the rolling constraint and shared wheel rotation, but the amplitudes remain bounded with no systematic growth or decay. This is consistent with energy conservation for a system described by a time-independent Lagrangian.

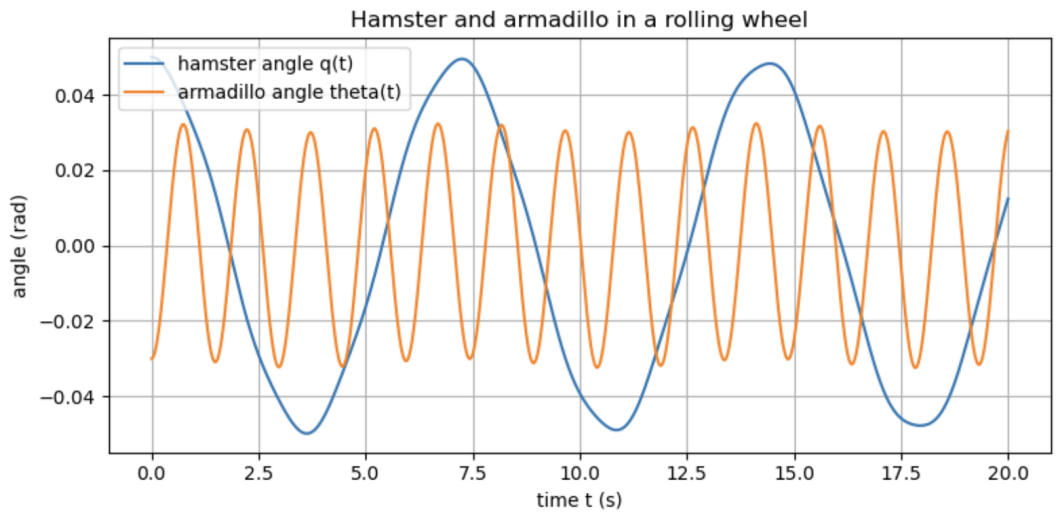

Figure 2: Numerical solution for $q(t)$ and $\theta(t)$ over $0 \leq t \leq 20$ s.

How much "directing" ChatGPT needed

The project statement notes that a generative AI will most likely not be able to solve the problem directly. This was true in my experience.

- **Coordinates and constraints:** When I provided only the problem statement, ChatGPT attempted to set up the dynamics but chose coordinate conventions different from the textbook and initially omitted the rotational kinetic energy of the ball. It also wrote the no-slip constraint incorrectly. After consulting the book solution, I explicitly supplied

  $$h\dot{\psi} + (R - h)\dot{\theta} = R\dot{\varphi}$$

  and required the AI to use this relation consistently.

- **Time-dependent vs. time-independent Lagrangian:** ChatGPT initially retained explicit time dependence from the hamster motion. I had to direct it to introduce $q = \varphi + \Omega_H t$ so that the Lagrangian becomes time-independent (as in the book), enabling a clean energy-conservation interpretation.

- **Algebra and code:** When asked for the equations of motion, ChatGPT's hand-expanded equations were long and error-prone. I redirected it to encode the Lagrangian symbolically in SymPy, compute Euler–Lagrange equations by differentiation, and solve for $\ddot{q}$ and $\ddot{\theta}$ automatically.

Overall, the AI was useful as a computational and coding assistant, but only after the physical model, constraints, and coordinate choices were carefully specified and repeatedly checked against the book's solution.